Exercise sheet: Week 7 - solutions

1. $\underline{X}$ is a random sample from $\mathrm{Poi}(\lambda)$. Suppose we choose as prior $\lambda \sim \mathrm{Gamma}(\alpha, \beta)$. Then, given a random sample $\underline{x}$ we have

$$\begin{aligned} \pi(\lambda|\underline{x}) &\propto \pi(\lambda) L(\lambda; \underline{x}) \\ &\propto \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \frac{1}{\Pi\, x_i!} \lambda^{\sum x_i} e^{-n\lambda}, \quad \text{where } n \text{ is the sample size,} \\ &\propto \lambda^{\alpha+\sum x_i - 1} e^{-(\beta+n)\lambda}, \end{aligned} \tag{1}$$

   implying $\lambda|\underline{x} \sim \mathrm{Gamma}(\alpha + \sum x_i,\ \beta + n)$. This is the same family as the prior and therefore the Gamma distribution is the conjugate prior.

   Data: $n = 10$, $\sum x_i = 28$, $\lambda|\underline{x} \sim \mathrm{Gamma}(28.5, 10.5)$.
   This gives a posterior mean $= \frac{28.5}{10.5}$ and posterior variance $= \frac{28.5}{(10.5)^2}$.

2. Again using $\mathrm{Gamma}(0.5, 0.5)$ as the prior:
   (i) $\lambda \,\big|\, \sum_1^5 x_i = 12 \sim \mathrm{Gamma}(12.5, 5.5)$.

   (ii) Using $\mathrm{Gamma}(12.5, 5.5)$ as a prior for observation of $x_6, \ldots, x_{10}$ we obtain

$$\lambda \,\Bigg|\, \sum_6^{10} x_i = 16 \sim \mathrm{Gamma}(12.5 + 16, 5.5 + 5) \sim \mathrm{Gamma}(28.5, 10.5). \tag{2}$$

   (iii) The analysis required here is analogous to that of question 1.

3. Here $\underline{X}$ is a random sample from $\mathrm{Pareto}(\theta)$. For the likelihood we compute

$$\begin{aligned} L(\theta; \underline{x}) &= \prod_{i=1}^n \theta\,(1+x_i)^{-(\theta+1)} \\ &= \theta^n \Big(\prod_{i=1}^n (1+x_i) \Big)^{-(\theta+1)} \\ &= \theta^n \exp\Big(-(\theta+1) \log\Big(\prod_{i=1}^n (1+x_i) \Big) \Big) \\ &\propto \theta^n \exp\big(-\theta\, t(\underline{x}) \big), \end{aligned} \tag{3}$$

   where $t(\underline{x}) = \sum_{i=1}^n \log(1+x_i)$. (NB.: $t(\underline{x})$ is sufficient.)
   If the prior for θ is $\mathrm{Gamma}(\alpha, \beta)$, then $\pi(\theta|\underline{x})$ is $\mathrm{Gamma}(\alpha + n, \beta + t(\underline{x}))$.

4. This time $\underline{X}$ is a random sample from $\mathrm{U}(0,\theta)$. Recall from tutorial sheets 1 and 2 that

$$L(\theta;\underline{x}) = \begin{cases} \prod_{i=1}^{n} \frac{1}{\theta}, & \text{if } 0 < x_i < \theta,\ \forall\, i \\ 0, & \text{otherwise,} \end{cases} \tag{4}$$

such that $L(\theta;\underline{x}) = \frac{1}{\theta^n}$ for $\theta > x_{(n)}$ and zero otherwise.

Suppose we choose the prior from a given Pareto family

$$\pi(\theta) = \frac{\beta\,\alpha^{\beta}}{\theta^{\beta+1}}, \quad \theta > \alpha. \tag{5}$$

Then,

$$\begin{aligned} \pi(\theta|\underline{x}) \;&\propto\; \pi(\theta)L(\theta;\underline{x}) \\ &= \frac{\beta\,\alpha^{\beta}}{\theta^{\beta+1}} \cdot \frac{1}{\theta^{n}} \quad \text{if } \theta > \max\{\alpha, x_{(n)}\} \\ &\propto\; \frac{1}{\theta^{n+\beta+1}} \quad \text{under the same condition.} \end{aligned} \tag{6}$$

Therefore the posterior is from the same family, so that the above Pareto family is the conjugate prior density.

5. **Prior:** Given $\mu \sim N(\alpha, \beta)$, the prior density is

$$\pi(\mu) = \frac{1}{\sqrt{2\pi\beta}} \exp\left\{-\frac{1}{2\beta}(\mu-\alpha)^2\right\},$$

Likelihood: From $X_i \sim N(\mu, \sigma^2)$, with σ known. Then the likelihood function is

$$L(\mu;\underline{x}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_i-\mu)^2\right) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right).$$

Completing the square gives

$$L(\mu;\underline{x}) \propto \exp\left(-\frac{n}{2\sigma^2}(\bar{x}-\mu)^2\right),$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is the sample mean.

Posterior: By Bayes' theorem,

$$\pi(\mu \mid y) \propto \exp\left\{-\frac{n}{2\sigma^2}(\bar{x}-\mu)^2 - \frac{1}{2\beta}(\mu-\alpha)^2\right\}.$$

Collecting terms in μ, the posterior is Normal:

$$\pi(\mu \mid y) \sim N(\mu_n, V_n),$$

where the posterior mean and variance are

$$\mu_n = \frac{\frac{\alpha}{\beta} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{\beta} + \frac{n}{\sigma^2}}, \qquad V_n = \left(\frac{1}{\beta} + \frac{n}{\sigma^2}\right)^{-1}$$

Weighted average form. Equivalently,

$$\mu_n = \frac{n\beta\bar{x} + \sigma^2\alpha}{n\beta + \sigma^2} = w\bar{x} + (1-w)\alpha, \quad w = \frac{n\beta}{n\beta + \sigma^2}.$$

Comment: The posterior mean is a weighted average of the sample mean $\bar{x}$ and the prior mean α.

- As n increases, more weight is placed on $\bar{x}$; in the limit $n \to \infty$, $\mu_n \to \bar{x}$ and the data dominate.
- As β decreases, more weight is placed on α; in the limit $\beta \to 0$, $\mu_n \to \alpha$ and the prior dominates.
- This illustrates the Bayesian principle of compromise between prior belief and data.